The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

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1 Introduction

Some patterns are hard to turn a blind eye on, especially when they might sway the outcome one way or another. While playing my favourite card game ”Speed”, I noticed that the more rounds of a game we played, the more ordered the cards became, which made me wonder if this was in fact true, or if I had jumped to logical conclusions and if ... Playing cards is one of my favourite ways to connect with people, and so I wanted to explore the maths behind this game and work out if my assumption proved true.

GAME EXPLAINED

In the game of ”Speed”, the deck is split evenly in half, and each player sets up their half of the deck, respectively, in a layered formation, as shown below:

Figure 1: first layer - one faced up, four faced down

Figure 2: second layer - one faced up, three faced down

Figure 3: third layer - one faced up, two faced down

Figure 4: fourth layer - one faced up, one faced down

Figure 5: fifth layer - one faced up

The remainder of their half deck is placed in the centre, and both piles now act as draw piles, that are played alongside onto the playing pile:

Figure 6: final setup of Speed

2 Exploration

With the aim of understanding how the number of rounds affect the orderedness of the cards in play, I experimented with collecting data on, and modelling the following relationship:

x-axis: the number of rounds played (independent variable) y-axis: the cards flipped from the draw pile (dependent variable)

I first used the variable of cards flipped from the draw pile because this is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck, reducing the scope and therefore increasing the validity. However, after collecting this data and graphing it on Desmos, I discovered that the relationship gave a random result that didn't indicate any kind of trend, like the one I had hypothesised.

Following this setback, I tried a different approach that still assessed the number of rounds as the independent variable, however changed the dependent variable to include the whole deck of cards, rather than just the few cards in play, faced up in front of each player. Additionally, I changed the actual determining factor to be the longest run that exists in each player's hand, rather than the number of cards drawn, to more accurately represent the orderedness of cards as it allows for cards in order, both up and down, to be recorded, rather than the indicator of cards drawn which can get skewed by the cards that players have available.

In this new exploration, I used the following factors:

Independent variable: number of the round (n) Dependent variable: length of the longest chain (c)

Data collection:

I played 10 games of Speed, limiting each game to 5 rounds.

I took count of the c value before each round, and once after the final, fifth round, producing 6 data points for each game.

After collecting adequate data on these variables, I found the average ratio (r) per player (r1 and r2), for each round from 0-5. (Be explicit about mistakes, new decisions, disregarding a certain approach etc.)

abc bc can be grouped d can be asymptote